# Database Project – Phase 1

**Course Name: Database Systems Design**

**Project: Gym Membership System**

**Team Number : 27**

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# Database Project – Phase 1 Report

## 1. Introduction

This project presents the design and implementation of a Gym Management System database.
The system is designed to handle daily gym operations including managing members, staff, class scheduling,
spa services, bookings, and financial transactions.

The primary goal of this database is to ensure efficient data organization, eliminate redundancy, and maintain data integrity. Additionally, the system is designed to be scalable, allowing future expansion
such as adding new services, tracking performance metrics, and integrating with external systems.

## 2. Entity Descriptions

The database is divided into multiple entities, each serving a specific role in the system:

Member: Represents gym clients and stores personal information such as name, email, phone number, and join date.
Each member is associated with a membership plan.

Membership_Plan: Defines different subscription types available in the gym, such as Basic, Standard, and VIP plans.
It also stores pricing information for each plan.

Staff: Contains employee data including their roles such as trainers, instructors, and administrative staff.

Class_Catalog: Stores general information about available gym classes, including class name, capacity, and duration.

Spa_Service: Contains details about spa services such as service name, duration, and additional cost.

Class_Session: Represents scheduled instances of classes, including date, time, and assigned instructor.

Class_Roster: Tracks the participation of members in class sessions and records attendance status.

Spa_Booking: Stores information about spa appointments linking members, services, and staff.

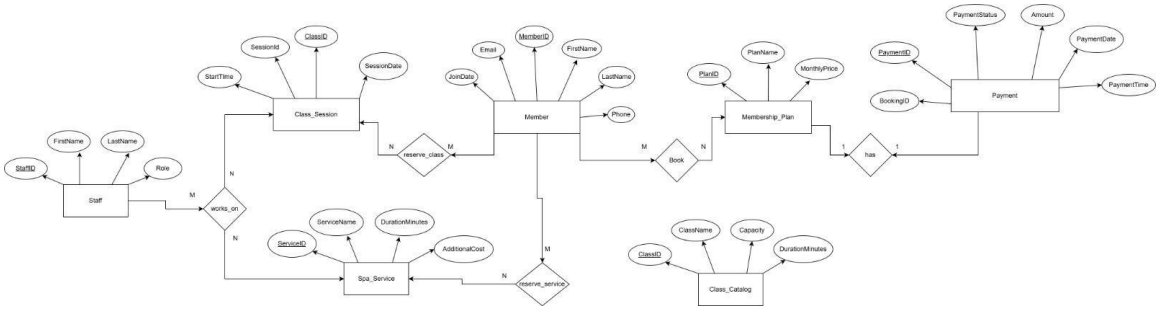
Payment: Records all financial transactions and links them to membership plans, bookings, or class attendance.

## 3. Business Rules

The system follows a set of business rules to maintain consistency and integrity:

- Each member must subscribe to exactly one membership plan.
- A membership plan can be assigned to multiple members.
- Each class can have multiple sessions scheduled over time.
- Each session is conducted by one staff member, while staff can conduct multiple sessions.
- Members can attend multiple sessions, and sessions can include multiple members.
- This many-to-many relationship is resolved using the Class_Roster table.
- Members can book multiple spa services on different dates.
- Each booking links one member, one service, and one staff member.
- Payments are always associated with a member and processed by a staff member.
- Payments may reference membership plans, spa bookings, or class attendance records.

## 4. ERD




## 5. Normalization

### 5.1 First Normal Form (1NF)

The database satisfies the First Normal Form as all tables contain atomic values. Each attribute holds a single value, and there are no repeating groups or multivalued attributes. Each table has a clearly defined primary key, ensuring that each record can be uniquely identified.

### 5.2 Second Normal Form (2NF)

The database satisfies the Second Normal Form as all non-key attributes are fully dependent on the entire primary key. Partial dependencies have been removed.

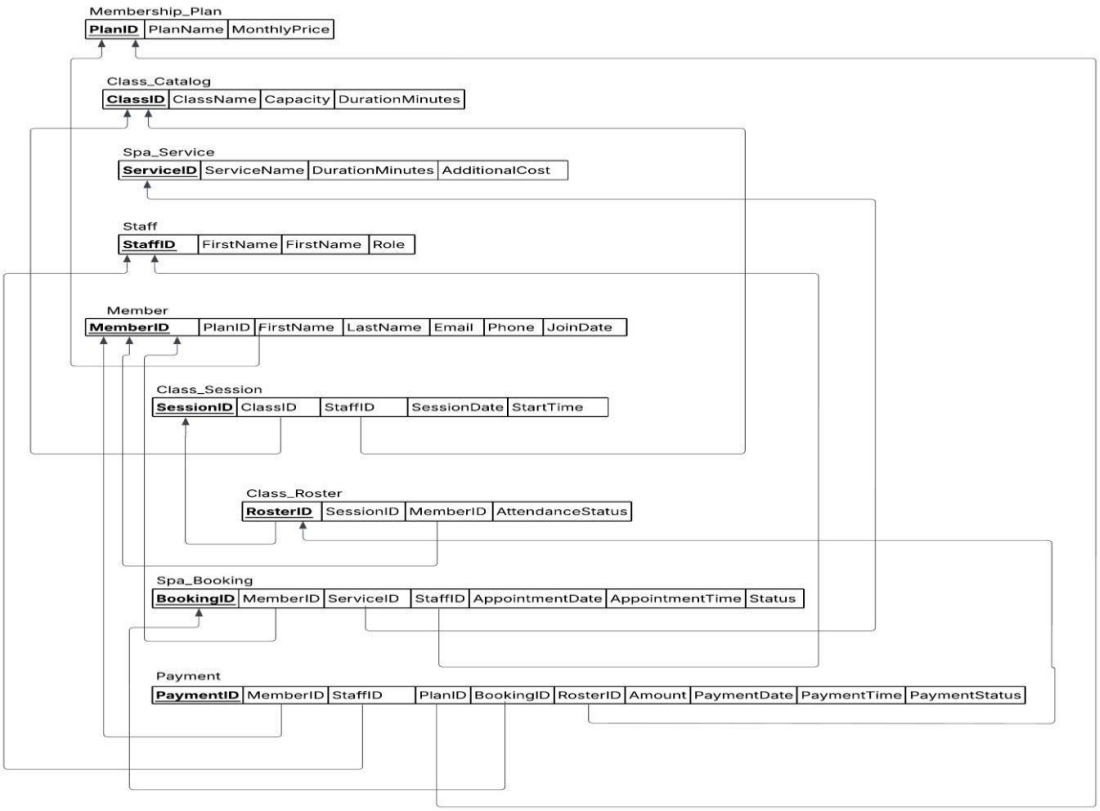
Additionally, many-to-many relationships are resolved using associative tables such as Class_Roster
and Spa_Booking, ensuring that data is stored efficiently and without duplication.

### 5.3 Third Normal Form (3NF)

The database satisfies the Third Normal Form as there are no transitive dependencies. All attributes depend only on the primary key of their respective tables.

For example, pricing details are stored only in Membership_Plan, and class details are stored in Class_Catalog.
This ensures that data redundancy is minimized and consistency is maintained across the database.

## 6. Relational Schema




The relational schema represents the database structure after converting the ERD into tables:

Membership_Plan(PlanID, PlanName, MonthlyPrice)
Member(MemberID, PlanID, FirstName, LastName, Email, Phone, JoinDate) Staff(StaffID, FirstName, LastName, Role)
Class_Catalog(ClassID, ClassName, Capacity, DurationMinutes) Class_Session(SessionID, ClassID, StaffID, SessionDate, StartTime) Class_Roster(RosterID, SessionID, MemberID, AttendanceStatus) Spa_Service(ServiceID, ServiceName, DurationMinutes, AdditionalCost) Spa_Booking(BookingID, MemberID, ServiceID, StaffID, AppointmentDate, AppointmentTime, Status)
Payment(PaymentID, MemberID, StaffID, PlanID, BookingID, RosterID, Amount, PaymentDate, PaymentTime, PaymentStatus)

## 7. Design Decisions

Several important design decisions were made to improve the system:

- Separation of entities to avoid redundancy and ensure clarity.
- Use of foreign keys to maintain relationships between tables.
- Use of associative tables to resolve many-to-many relationships.
- Proper normalization to improve performance and maintain consistency.

## 8. Conclusion

The database design is fully normalized up to Third Normal Form (3NF).
It ensures efficient data storage, minimizes redundancy, and maintains strong data integrity.

The system is flexible and scalable, making it suitable for real-world gym management applications.
Future improvements may include adding reporting features, analytics, and user interfaces.